EDA

Anni and Daniel

```r
library(tidyverse)
library(skimr)
library(knitr)
library(kableExtra)
library(scales)
```

```r
setwd("/Users/annieyu/math244_project")
wage_data <- read_csv("data/wage_data_raw.csv")
```

Exploring the Data

One of our outcome variable which we created is wage_gap, calcualted as the dollar difference between male and female median earnings. This variable captures the raw income disparity and allows us to examine the magnitude of the earnings gap. We expect this value to be generally positive, indicating men have a higher median earnings across most occupations.

A second outcome variable we are using, wage_percent_of_male, is included in our dataset. It measures female median earnings as a percentage of male median earnings. A value of 100 would indicate perfect parity; values below 100 indicate women earn less than men. This variable allows us to analyze relative disparity. We expect this variable to generally fall below 100 for most occupations.

Our key explanatory variables are:

-percent_female: The percent of females for specific occupation

-total_workers: Total estimated full-time workers > 16 years old

-major_category: Broad category of occupation

-minor_category: Fine category of occupation

-occupation: Specific job/career

-year: years from 2013-2016

```r
#Select the variables we plan to use
wages_selected <- wage_data %>%
  select(wage_percent_of_male, percent_female,
         year, major_category, minor_category,
         percent_female, total_workers,
         major_category, minor_category,
         occupation, year)
```

```r
# Summary table of numeric variables (except year)
wages_selected %>%
  select(where(is.numeric), -year) %>%
  summarise(across(everything(),
    list(
      N = ~sum(!is.na(.)),
      Mean = ~round(mean(., na.rm = TRUE), 1),
      SD = ~round(sd(., na.rm = TRUE), 1),
      Min = ~round(min(., na.rm = TRUE), 1),
      Median = ~round(median(., na.rm = TRUE), 1),
      Max = ~round(max(., na.rm = TRUE), 1)
    )
  )) %>%
  pivot_longer(
    everything(),
    names_to = c("Variable", "Statistic"),
    names_pattern = "(.*)_(.*)"
  ) %>%
  pivot_wider(names_from = Statistic, values_from = value) %>%
  kable(caption = "Table 1: Summary Statistics for Key Variables") %>%
  kable_styling(bootstrap_options = "striped", full_width = FALSE)
```

Table 1: Table 1: Summary Statistics for Key Variables

Variable	N	Mean	SD	Min	Median	Max
wage_percent_of_male	1242	84.0	9.4	50.9	85.2	117.4
percent_female	2088	36.0	27.5	0.0	32.4	100.0
total_workers	2088	196054.9	375361.0	658.0	58997.0	3758629.0

Data Wrangling and Transformation

```r
#Create a new variable: wage_gap
wage_data <- wage_data %>%
  mutate(wage_gap = total_earnings_male - total_earnings_female)
```

```r
# Cleaned dataset version 1: Dropped occupations with any missing wage_percent_of_male values

wage_data_complete_cases <- wage_data %>%
  group_by(occupation) %>%
  filter(!any(is.na(wage_percent_of_male))) %>%
  ungroup()

setwd("/Users/annieyu/math244_project")
write_csv(wage_data_complete_cases, "data/wage_data_complete_cases.csv")
```

```r
# Cleaned dataset version 2: Retained all observations, computed missing wage_percent_of_male vlau

wage_data_imputed <- wage_data %>%
  mutate(
    wage_percent_of_male = ifelse(
      is.na(wage_percent_of_male),
      (total_earnings_female / total_earnings_male) * 100,
      wage_percent_of_male
    )
  )

setwd("/Users/annieyu/math244_project")
# Save dataset
write_csv(wage_data_imputed, "data/wage_data_imputed.csv")
```

We saw a considerable amount of missing values in the wage_percent_of_male column of our dataset. The author of TidyTuesday noted this is because those occupations have small sample sizes in total_workers, workers_male, and workers_female. The wage_percent_of_male metric calculated from those values may be unrepresentative and was thus reported as NA.

To address this, we conducted two separate analyses. In one version, we drop all observation with NA wage_percent_of_male values, and also filter out all observation of that same occupation across other years for better reliability and consistency for our time-series analysis.

In the other, we retained these observations, as occupations with small workforce sizes or extreme gender composition may actually capture insightful patterns. For this set of analysis, we recomputed the NA wage_percent_of_male values using the total_earnings_male and total_earnings_female columns.

In terms of data transformation, we created a new variable: wage_gap. It measures the dollar difference between the male and female median earnings (male minus female). This variable captures the magnitude of the wage disparity in absolute terms and complements the percentage-based measure.

Codebook

```r
codebook <- tribble(
  ~variable, ~class, ~description,
  "year", "integer", "Year",
  "occupation", "character", "Specific job/career",
  "major_category", "character", "Broad category of occupation",
  "minor_category", "character", "Fine category of occupation",
  "total_workers", "double", "Total estimated full-time workers > 16 years old",
  "workers_male", "double", "Estimated MALE full-time workers > 16 years old",
  "workers_female", "double", "Estimated FEMALE full-time workers > 16 years old",
  "percent_female", "double", "Percent of females in occupation",
  "total_earnings", "double", "Median earnings for all workers",
  "total_earnings_male", "double", "Median earnings for male workers",
  "total_earnings_female", "double", "Median earnings for female workers",
  "wage_percent_of_male", "double", "Female earnings as % of male earnings",
  "wage_gap", "double", "Difference between male and female median earnings"
)
```

```r
codebook %>%
  kable(caption = "Data Dictionary") %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))
```

Table 2: Data Dictionary

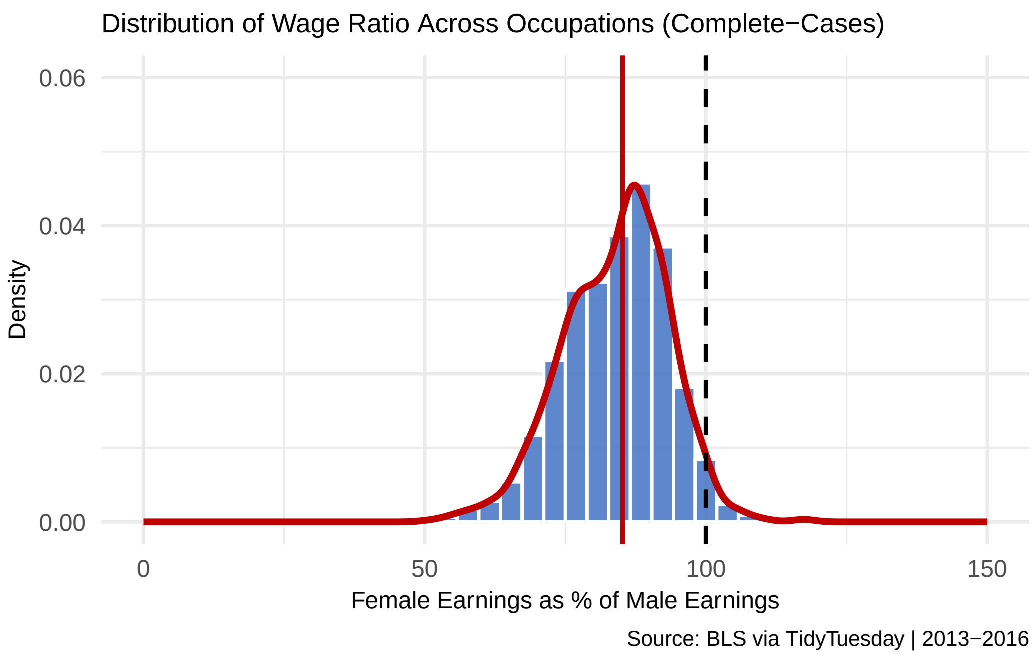
variable	class	description
year	integer	Year
occupation	character	Specific job/career
major_category	character	Broad category of occupation
minor_category	character	Fine category of occupation
total_workers	double	Total estimated full-time workers > 16 years old
workers_male	double	Estimated MALE full-time workers > 16 years old
workers_female	double	Estimated FEMALE full-time workers > 16 years old
percent_female	double	Percent of females in occupation
total_earnings	double	Median earnings for all workers
total_earnings_male	double	Median earnings for male workers
total_earnings_female	double	Median earnings for female workers
wage_percent_of_male	double	Female earnings as % of male earnings
wage_gap	double	Difference between male and female median earnings

Data Visualization

```r
# Plot 1: Distribution of wage_percent_of_male (Complete-Cases)

ggplot(wage_data_complete_cases, aes(x = wage_percent_of_male)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 40,
                 fill = "#4472C4",
                 color = "white",
                 alpha = 0.85) +
  geom_density(color = "#C00000", linewidth = 1.2) +
  geom_vline(xintercept = 100, linetype = "dashed",
             color = "black", linewidth = 0.8) +
  geom_vline(xintercept = median(wage_data_complete_cases$wage_percent_of_male, na.rm = TRUE),
             color = "#C00000", linewidth = 0.8) +
  scale_x_continuous(limits = c(0, 150)) +
  scale_y_continuous(limits = c(0, 0.06)) +
  labs(
    title = "Distribution of Wage Ratio Across Occupations (Complete-Cases)",
    x = "Female Earnings as % of Male Earnings",
    y = "Density",
    caption = "Source: BLS via TidyTuesday | 2013-2016"
  ) +
  theme_minimal(base_size = 13)+
theme(
```

```r
  plot.title = element_text(size = 10),
  axis.text = element_text(size = 9),
  axis.title = element_text(size = 9),
  plot.caption = element_text(size = 8)
)
```



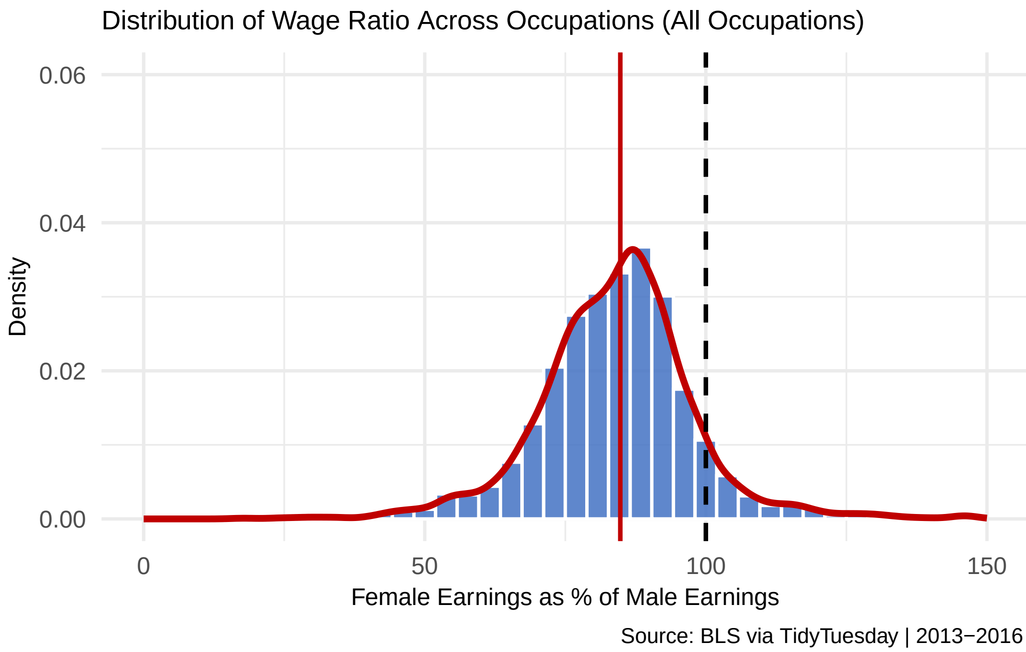

```r
setwd("/Users/annieyu/math244_project")

# Save the figure
ggsave("figures/plot1_wage_ratio_distribution_complete_cases.png", width = 7, height = 5, dpi = 30
```

```r
# Plot 2: Distribution of wage_percent_of_male (All Occupations)

ggplot(wage_data_imputed, aes(x = wage_percent_of_male)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 40,
                 fill = "#4472C4",
                 color = "white",
                 alpha = 0.85) +
  geom_density(color = "#C00000", linewidth = 1.2) +
  geom_vline(xintercept = 100, linetype = "dashed",
             color = "black", linewidth = 0.8) +
  geom_vline(xintercept = median(wage_data_imputed$wage_percent_of_male, na.rm = TRUE),
             color = "#C00000", linewidth = 0.8) +
  scale_x_continuous(limits = c(0, 150)) +
  scale_y_continuous(limits = c(0, 0.06)) +
  labs(
    title = "Distribution of Wage Ratio Across Occupations (All Occupations)",
    x = "Female Earnings as % of Male Earnings",
    y = "Density",
```

```
    caption = "Source: BLS via TidyTuesday | 2013-2016 "
  ) +
  theme_minimal(base_size = 13)+
theme(
  plot.title = element_text(size = 10),
  axis.text = element_text(size = 9),
  axis.title = element_text(size = 9),
  plot.caption = element_text(size = 8)
)
```



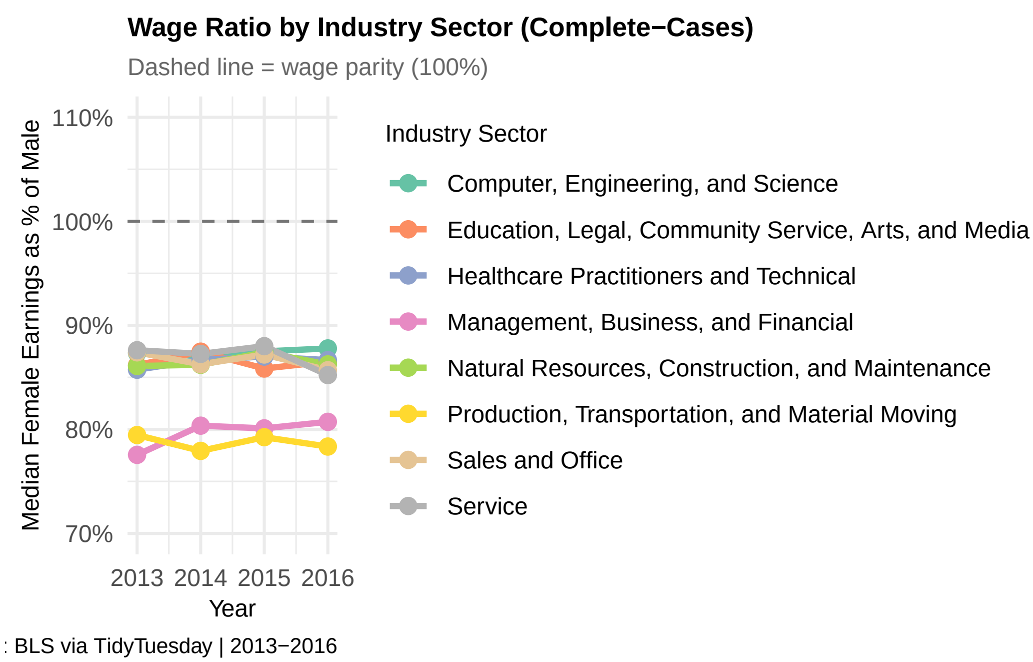

```
setwd("/Users/annieyu/math244_project")

# Save the figure
ggsave("figures/plot2_wage_ratio_distribution_imputed.png", width = 7, height = 5, dpi = 300)
```

```
# Plot 3: Trend of median wage parity over time, by major_category (Complete-Cases)

wage_data_complete_cases %>%
  group_by(year, major_category) %>%
  summarise(median_wage_pct = median(wage_percent_of_male, na.rm = TRUE),
            .groups = "drop") %>%
  ggplot(aes(x = year, y = median_wage_pct,
             color = major_category, group = major_category)) +
  geom_line(linewidth = 1.1) +
  geom_point(size = 2.5) +
  geom_hline(yintercept = 100, linetype = "dashed",
             color = "black", alpha = 0.5) +
  scale_x_continuous(breaks = 2013:2016) +
  scale_y_continuous(labels = function(x) paste0(x, "%"),
                     limits = c(70, 110)) +
```

```r
  scale_color_brewer(palette = "Set2") +
  labs(
    title    = "Wage Ratio by Industry Sector (Complete-Cases)",
    subtitle = "Dashed line = wage parity (100%)",
    x        = "Year",
    y        = "Median Female Earnings as % of Male",
    color    = "Industry Sector",
    caption  = "Source: BLS via TidyTuesday | 2013-2016"
  ) +
 theme_minimal(base_size = 12) +
  theme(
    plot.title      = element_text(face = "bold", size = 10),
    plot.subtitle   = element_text(size = 9, color = "gray40"),
    axis.text       = element_text(size = 9),
    axis.title      = element_text(size = 9),
    legend.text     = element_text(size = 9),
    legend.title    = element_text(size = 9),
    plot.caption    = element_text(size = 8),
    legend.position = "right"
  )
```



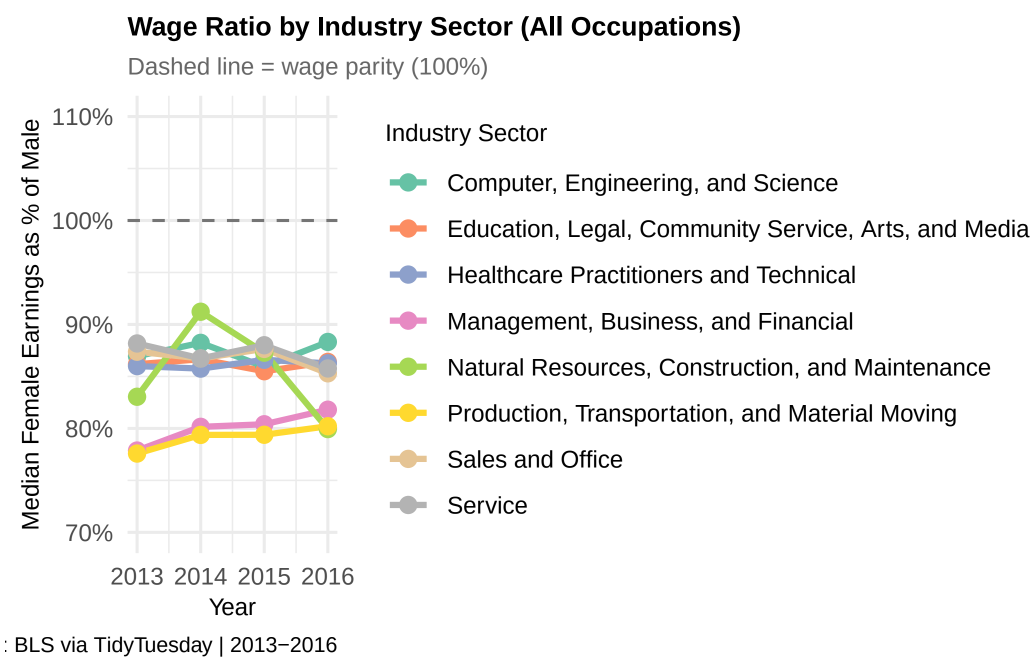

```r
setwd("/Users/annieyu/math244_project")

# Save the figure
ggsave("figures/plot3_wage_ratio_trend_by_sector_complete_cases.png", width = 7, height = 6, dpi =
```

```r
# Plot 4: Trend of median wage parity over time, by major_category (All Occupations)
# Outcome variable (wage_percent_of_male) x time x category

wage_data_imputed %>%
```

```r
group_by(year, major_category) %>%
summarise(median_wage_pct = median(wage_percent_of_male, na.rm = TRUE),
          .groups = "drop") %>%
ggplot(aes(x = year, y = median_wage_pct,
           color = major_category, group = major_category)) +
geom_line(linewidth = 1.1) +
geom_point(size = 2.5) +
geom_hline(yintercept = 100, linetype = "dashed",
           color = "black", alpha = 0.5) +
scale_x_continuous(breaks = 2013:2016) +
scale_y_continuous(labels = function(x) paste0(x, "%"),
                   limits = c(70, 110)) +
scale_color_brewer(palette = "Set2") +
labs(
  title    = "Wage Ratio by Industry Sector (All Occupations)",
  subtitle = "Dashed line = wage parity (100%)",
  x        = "Year",
  y        = "Median Female Earnings as % of Male",
  color    = "Industry Sector",
  caption  = "Source: BLS via TidyTuesday | 2013-2016"
) +
theme_minimal(base_size = 12) +
theme(
  plot.title     = element_text(face = "bold", size = 10),
  plot.subtitle  = element_text(size = 9, color = "gray40"),
  axis.text      = element_text(size = 9),
  axis.title     = element_text(size = 9),
  legend.text    = element_text(size = 9),
  legend.title   = element_text(size = 9),
  plot.caption   = element_text(size = 8),
  legend.position = "right"
)
```



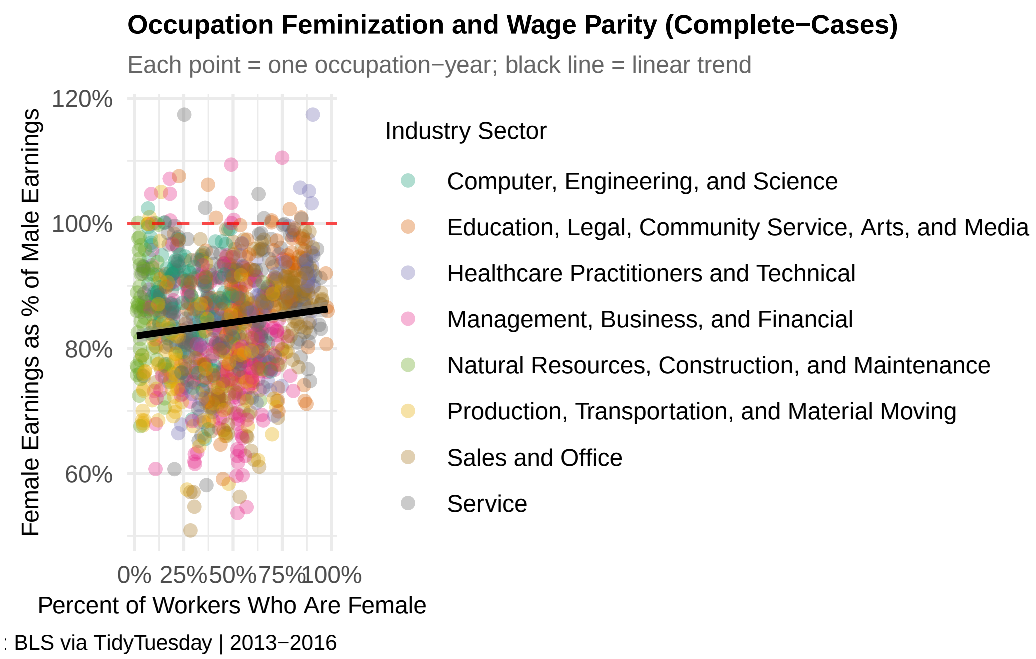

```
setwd("/Users/annieyu/math244_project")

# Save the figure
ggsave("figures/plot4_wage_ratio_trend_by_sector_imputed.png", width = 7, height = 6, dpi = 300)
```

```
# Plot 5: Scatter - percent_female vs wage_percent_of_male (Complete-Cases)

ggplot(wage_data_complete_cases, aes(x = percent_female, y = wage_percent_of_male,
                                     color = major_category)) +
  geom_point(alpha = 0.35, size = 1.8) +
  geom_smooth(method = "lm", se = TRUE,
              color = "black", linewidth = 1.2,
              linetype = "solid") +
  geom_hline(yintercept = 100, linetype = "dashed",
             color = "red", alpha = 0.7) +
  scale_x_continuous(labels = function(x) paste0(x, "%")) +
  scale_y_continuous(labels = function(x) paste0(x, "%")) +
  scale_color_brewer(palette = "Dark2") +
  labs(
    title    = "Occupation Feminization and Wage Parity (Complete-Cases)",
    subtitle = "Each point = one occupation-year; black line = linear trend",
    x        = "Percent of Workers Who Are Female",
    y        = "Female Earnings as % of Male Earnings",
    color    = "Industry Sector",
    caption  = "Source: BLS via TidyTuesday | 2013-2016"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title     = element_text(face = "bold", size = 10),
    plot.subtitle  = element_text(size = 9, color = "gray40"),
    axis.text      = element_text(size = 9),
```

```
    axis.title     = element_text(size = 9),
    legend.text    = element_text(size = 9),
    legend.title   = element_text(size = 9),
    plot.caption   = element_text(size = 8),
    legend.position = "right"
  )
```



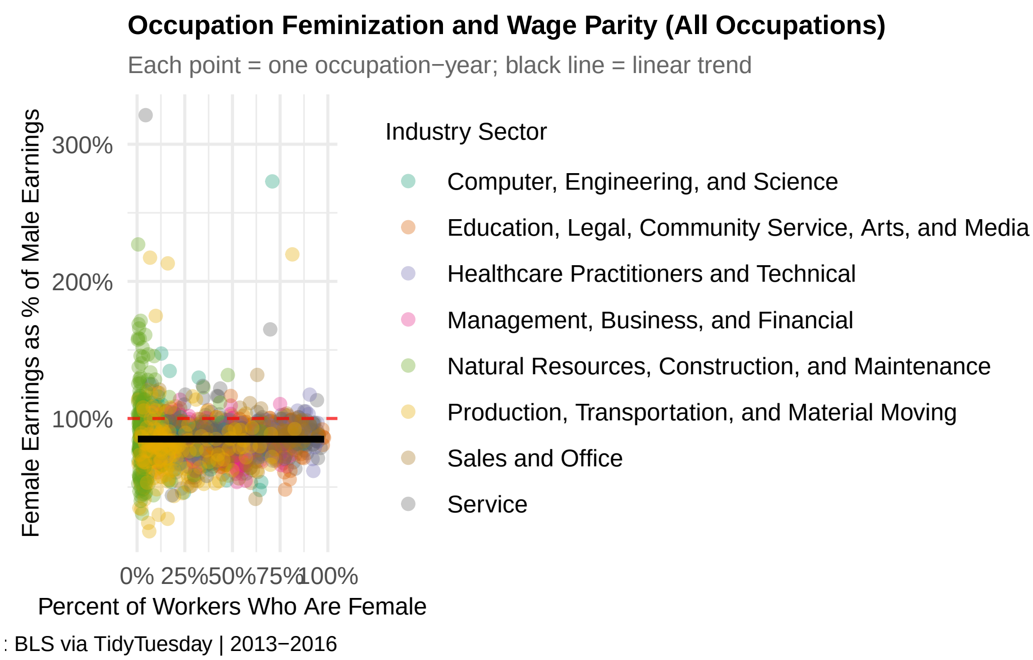

```
setwd("/Users/annieyu/math244_project")

# Save the figure
ggsave("figures/plot5_feminization_scatter_complete_cases.png", width = 7, height = 6, dpi = 300)
```

```
# Plot 6: Scatter - percent_female vs wage_percent_of_male (All Occupations)

ggplot(wage_data_imputed, aes(x = percent_female, y = wage_percent_of_male,
                                     color = major_category)) +
  geom_point(alpha = 0.35, size = 1.8) +
  geom_smooth(method = "lm", se = TRUE,
              color = "black", linewidth = 1.2,
              linetype = "solid") +
  geom_hline(yintercept = 100, linetype = "dashed",
             color = "red", alpha = 0.7) +
  scale_x_continuous(labels = function(x) paste0(x, "%")) +
  scale_y_continuous(labels = function(x) paste0(x, "%")) +
  scale_color_brewer(palette = "Dark2") +
  labs(
    title    = "Occupation Feminization and Wage Parity (All Occupations)",
    subtitle = "Each point = one occupation-year; black line = linear trend",
    x        = "Percent of Workers Who Are Female",
    y        = "Female Earnings as % of Male Earnings",
```

```
    color    = "Industry Sector",
    caption  = "Source: BLS via TidyTuesday | 2013-2016"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title     = element_text(face = "bold", size = 10),
    plot.subtitle  = element_text(size = 9, color = "gray40"),
    axis.text      = element_text(size = 9),
    axis.title     = element_text(size = 9),
    legend.text    = element_text(size = 9),
    legend.title   = element_text(size = 9),
    plot.caption   = element_text(size = 8),
    legend.position = "right"
  )
```




```
setwd("/Users/annieyu/math244_project")

# Save the figure
ggsave("figures/plot6_feminization_scatter_imputed.png", width = 7, height = 6, dpi = 300)
```

Data Analysis

Next, we can apply machine learning techniques, such as PCA, to reveal underlying patterns across occupations. Clustering methods such as k-means could be used to group occupations into categories based on similarities in wage structures and gender composition. It can help identify whether there is a better way to cluster occupations than the given major and minor categories we have. We'll also use regression models with interaction terms, which can help us examine relationships between gender composition and wage differences across industries.